



Indian Institute of Technology (Indian School of Mines) Dhanbad
Department of Computer Science and Engineering
Principles of Artificial Intelligence (CSO303)
End Semester Exam, Date: November 21, 2024

Timing: 09:00 AM to 12:00 PM Monsoon Semester (2024-2025) Maximum marks: 100

1. (a) Compare and contrast the sigmoid, tanh, and ReLU activation functions in terms of their mathematical properties, advantages, disadvantages, and use cases.
- (b) Consider a multilayer perceptron with one hidden layer using the sigmoid activation function. Derive the gradient of the loss function with respect to the weights of: i. The output layer and ii. The hidden layer. Explain the significance of the chain rule in this derivation.
- (c) Discuss the limitations of the minimax algorithm in large-scale games like chess. How can techniques like heuristic evaluation functions and iterative deepening be integrated to make minimax practical for such complex games? Provide an example.

(6+6+3)

2. (a) Explain the difference between constraint-based and score-based approaches for learning the structure of Bayesian Networks. Provide one advantage and one disadvantage of each approach.
- (b) Define the concept of "premature convergence" in Genetic Algorithms. Propose two strategies to mitigate this issue.
- (c) GAs are often criticized for their high computational cost. Explain why this is the case and suggest at least three ways to reduce the computational burden without compromising solution quality.

(5+5+5)

3. (a) You are given a map-colouring CSP with eight regions: A, B, C, D, E, F, G, and H. Each region must be coloured using one of three colours: Red, Green, or Blue. The adjacency constraints are as follows:
 - i. A is adjacent to B, C, D.
 - ii. B is adjacent to A, E, F.
 - iii. C is adjacent to A, E, G.
 - iv. D is adjacent to A, F, H.
 - v. E is adjacent to B, C, G.
 - vi. F is adjacent to B, D, H.
 - vii. G is adjacent to C, E, H.
 - viii. H is adjacent to D, F, G.Solve this CSP using the Backtracking algorithm with Degree Heuristic for variable ordering. Show each step in the assignment process, including where backtracking occurs. Additionally, explain how using the Least Constraining Value (LCV) heuristic and Minimum Remaining Value (MRV) could affect the outcome and efficiency of the solution.
- (b) Consider a CSP with seven variables A, B, C, D, E, F, and G, each having a domain of [1,2,3,4,5]. The constraints are:
 - i. $A + B = C$
 - ii. $B < D$
 - iii. $C \neq E$
 - iv. $D + E = F$
 - v. $F > G$

vi. $G \neq A$

Solve this CSP using constraint propagation with arc consistency (AC-3) to reduce the domains of each variable. Show each step in the domain reduction process. After completing arc consistency, use backtracking to find a solution. Document each assignment and any backtracking steps, and discuss how arc consistency helped reduce the search space before backtracking.

(8+7)

4. (a) Consider the following first-order logic (FOL) sentences:

- i. $\forall x, y [P(x, y) \wedge Q(y, x) \Rightarrow R(x, y)]$
- ii. $\forall x, y [S(x, \text{Bob}) \wedge S(y, x) \Rightarrow P(x, y)]$
- iii. $\forall x, y [S(x, y) \Rightarrow Q(y, x)]$
- iv. $\forall x, y [T(x, y, x) \Rightarrow Q(x, y)]$
- v. $\forall x, y [T(x, x, y) \Rightarrow Q(x, y)]$

Facts:

$T(\text{Alice}, \text{Dawn}, \text{Alice})$
 $T(\text{Eve}, \text{Carl}, \text{Eve})$
 $T(\text{Alice}, \text{Bob}, \text{Dawn})$
 $T(\text{Carl}, \text{Carl}, \text{Alice})$
 $S(\text{Bob}, \text{Alice})$
 $S(\text{Carl}, \text{Bob})$
 $S(\text{Dawn}, \text{Carl})$
 $S(\text{Carl}, \text{Dawn})$
 $S(\text{Alice}, \text{Dawn})$
 $S(\text{Eve}, \text{Carl})$

Use backward chaining to find **ALL** answers for the following queries. When matching rules, proceed from top to bottom, and evaluate subgoals from left to right.

Queries:

- i. $\exists x Q(\text{Alice}, x)$ (Find all x such that $Q(\text{Alice}, x)$ holds.)
- ii. $\exists x, y R(x, y)$ (Find all pairs (x, y) such that $R(x, y)$ holds.)

(b) Determine whether the expressions p and q unify with each other in each of the following cases. If so, give the most general unifier; If not, give a brief explanation (Assume that the upper case letters are (object, predicate, or function) constants and that the lower case letters are variables).

- i. $p = F(G(v), H(u, v)); q = F(w, J(x, y))$
- ii. $p = F(x, F(u, x)); q = F(F(y, A), F(z, F(B, z)))$
- iii. $p = F(x1, G(x2, x3), x2, B); q = F(G(H(A, x5), x2), x1, H(A, x4), x4)$

(c) Convert the following English statements to statements in first order logic.

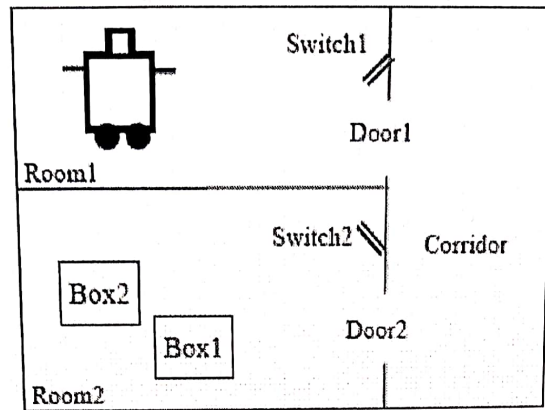
- i. every boy or girl is a child
- ii. every child gets a doll or a train or a lump of coal
- iii. no boy gets any doll
- iv. no child who is good gets any lump of coal
- v. Jack is a boy

Using the above five axioms construct a proof by resolution of the statement: "if Jack doesn't get a train, then Jack is not a good boy".

(5+5+5)

5. Consider the world of Shakey the robot, as shown below. Shakey has the following six actions available:

- i. $\text{Go}(x, y)$, which moves Shakey from x to y . It requires Shakey to be at x and that x and y are locations in the same room. By convention a door between two rooms is in both of them, and the corridor counts as a room.
- ii. $\text{Push}(b, x, y)$, which allows Shakey to push a box b from location x to location y . Both Shakey and the box must be at the same location before this action can be used.
- iii. $\text{ClimbUp}(b, x)$, which allows Shakey to climb onto box b at location x . Both Shakey



and the box must be at the same location before this action can be used. Also Shakey must be on the Floor.

iv. $\text{ClimbDown}(b,x)$, which allows Shakey to climb down from a box b at location x . Shakey must be on the box and the box must be in location x before this action can be used.

v. $\text{TurnOn}(s,x)$, which allows Shakey to turn on switch s which is located at location x . Shakey must be on top of a box at the switch's location before this action can be used.

vi. $\text{TurnOff}(s,x)$, which allows Shakey to turn off switch s which is located at location x . Shakey must be on top of a box at the switch's location before this action can be used.

(a) Using STRIPS syntax, define the six actions from above. In your action definitions, use only the following predicates: $\text{Box}(b)$ to mean that box b is a box; $\text{In}(x,r)$ to mean that location x is in room r ; $\text{At}(x,y)$ to mean that the object x is at location y ; $\text{ShakeyOn}(x)$ to mean that Shakey is on the object x ; $\text{Switch}(s)$ to mean that s is a switch; and $\text{SwitchOn}(s)$ to mean that the switch s is on. You may also use the constants Shakey and Floor in the action definitions.

(b) In the above problem, using STRIPS, define the initial state depicted. Use only the predicates from part (a) and the constants Box1, Box2, Switch1, Switch2, Floor, Shakey, Room1, Room2, Corridor, L_{Door1} , L_{Door2} , $L_{\text{ShakeyStart}}$, L_{Switch1} , $L_{\text{Box1Start}}$, $L_{\text{Box2Start}}$, L_{Switch2} . The L_x constants are intended to represent the locations of x .

(c) Provide a totally ordered plan for Shakey to turn off Switch2 using the actions and the initial state defined in (a) and (b).

(6+2+2)

6. (a) Discuss one advantage and one disadvantage of using distance-weighted k-NN compared to standard k-NN. Discuss how using Cosine Similarity as a distance metric in k-NN could affect classification outcomes when applied to high-dimensional, sparse data.
- (b) State the Naive Independence Assumption in Naive Bayes. Explain why this assumption might not hold in real-world datasets, with example. Discuss how the independence assumption affects the computational efficiency of Naive Bayes.
- (c) The following dataset helps predict whether a company employee will work remotely based on their position, department, and commute distance.

POSITION	DEPARTMENT	COMMUTE DISTANCE	WORKS REMOTELY?
✓ Engineer	IT	Long	Yes
✓ Manager	HR	Short	No
✓ Engineer	Finance	Medium	Yes
✓ Staff	IT	Short	No
✓ Manager	Finance	Long	Yes
✓ Staff	HR	Medium	No
✓ Engineer	Finance	Long	Yes
✓ Staff	IT	Short	No

Using entropy to guide your splitting decisions, construct a decision tree that predicts if an employee will work remotely. Show all calculations and each step of the decision-making process.

(4+3+8)

7. (a) Explain the difference between TF-IDF and word embeddings in text representation, giving an example of each. Discuss the advantages of contextual embeddings over traditional word embeddings. Identify one limitation of contextual embeddings and suggest a possible improvement.

- (b) Consider the story below:

Oliver had a busy day planned. He first stopped by his favorite café to grab a coffee. While waiting in line, he checked his watch and realized he was running late for his meeting at the office. He quickly paid for his coffee, took a sip, and headed out. As he reached the subway station, he found that the train was delayed due to maintenance. Frustrated, he called his assistant, Lisa, to inform her that he might be late and asked her to take notes in his absence. On the train, Oliver overheard a conversation about an upcoming art exhibition. Intrigued, he made a mental note to check it out later. When he finally arrived at the office, he thanked Lisa for covering for him and dove into his work.

Each inference below is based on a plausible interpretation of the story. For each inference, briefly explain whether that inference was primarily based on syntactic, semantic, pragmatic, discourse, or world knowledge. (Do not answer world knowledge unless none of the other categories are appropriate.) (2 marks for each inference)

- **Inference 1:** Oliver intended to get a coffee quickly because he was aware that he was running late for his meeting.
- **Inference 2:** When Oliver called Lisa to inform her that he would be late, he expected her to understand that she should take notes in his absence without him having to explain every detail.
- **Inference 3:** Oliver was likely interested in the art exhibition because he mentally noted to check it out later.
- **Inference 4:** The story implies that Oliver thanked Lisa specifically for taking notes and covering for him while he was late.
- **Inference 5:** Oliver's frustration at the delayed train suggests that he was relying on the subway to arrive at the office on time.

(5+10)

Semester: Monsoon

Examination: END SEM EXAM, B. Tech (Mining Engineering)

Subject: SURFACE MINING (MNC 300)

Instruction: Answer ALL questions

Session: 2024-25

Semester: V

Max. Marks: 100

Time: 3 Hours

Assume relevant data wherever necessary and mention them

Q.No.	Question	Marks
1.	Prepare a statement of heavy earth moving machinery required for a surface iron ore mine to be worked by shovel-dumper combination to produce 5.0 million tones of ore per annum with an average stripping ratio of 0.1 tonne/tonne. The mine has been planned to be worked with 12 m high benches. Average haul distance (one way) for both ore and waste is 1 km.	15
2.	Estimate the number of drills required to achieve a production of 10 million ton/year of coal from a surface mine having coal seam of thickness 10 m. The coal seam is flat and horizontal and the stripping ratio of the mine is $3m^3/ton$	15
3.	a) Derive the effective productivity of a loading shovel. b) In case of a shovel-dumper operation in a surface mine, prove that the effective productivity of shovel depends on the dumper capacity in case of single spotting condition only, but not in double spotting condition.	15
4.	a) How is the bucket capacity of a dragline estimated? b) Draw a balancing diagram for simple side casting operation of a dragline and derive the expression, in terms of known parameter, for i. height of spoil heap ii. reach of the dragline c) What is meant by MUF of a dragline and what is its physical significance?	15
5.	Explain with the help of suitable sketches the (i) full bench method, (ii) block mining method and (iii) step-cut method of continuous surface miner operation. Also mention the applicability and limitations of each of the methods. OR a) With the help of suitable sketches, describe lateral block cut method operation of bucket wheel excavator (BWE). b) Derive the theoretical hourly output for a bucket wheel excavator. c) Calculate the hourly productivity of the BWE from the following given data: i. Number of buckets on the wheel – 12 ii. Individual bucket capacity – 1400 lit iii. Wheel diameter – 10 m iv. Cutting speed at the knife edge – 2.2 m/sec v. % swelling of material to be excavated – 20	15
6.	a) Enlist and discuss the geometrical parameters of production benches in surface mines. How are these decided upon? b) Enlist and discuss the types of failure that may occur in highwall slopes in surface mines.	15
7.	Write short notes and (or) draw labeled diagram wherever necessary for the following a) Schematic presentation of mobile in-pit crushing and conveying system b) Concept of tipping load in pay-loaders	10

भारतीय प्रौद्योगिकी संस्थान (भारतीय खनि विद्यापीठ), धनबाद
Indian Institute of Technology (Indian School of Mines), Dhanbad
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

Mid-Semester Examination, Monsoon [2024-2025]

Date: 27-11-2024

Time: 3 hours

Full marks: 100

Course Name: Sociology of Health

Code: HSO404

All questions are compulsory

Answers should be written in bullets, with proper elaboration and appropriate examples.

1. Describe the arguments behind both 'favouring' and 'opposing' physician-assisted suicide. [10 + 10 = 20]
2. Discuss Pearlin's five types of role strains and Goldstein's five elements of alternative medicine. [10 + 10 = 20]
3. Discuss any five models of disability. [5 x 4 = 20]
4. Write on the history/evolution of Hospice and mention any five services provided in hospices. [5 + 5 = 10]
5. Discuss five models of physician-patient relationship [10]
6. Write on the following:
 - a. The Navajo Culture and advanced care directives [5]
 - b. Yin and Yang in healthcare [5]
 - c. Medicalization of ADHD [5]
 - d. Elaborate CAM, mention its two broad categories with examples. [1 + 4 = 5]

Assignment - Part
Mutual Pop





Indian Institute of Technology (ISM) Dhanbad
Department of Mathematics and Computing
MCC505 – Probability and Statistics

End Semester Exam

Total Marks: 48

Time: 3 hours.

Date: 25/11/2024

Instructions:

- Solve all questions. All the parts of each question should be solved at one place.
- Write your roll number, batch, group, course code, course name, etc. on your answer sheet.
- Calculator is allowed. Mobile/Smart watch/any other electronic item, etc. are not allowed in the exam hall.
- Mention the formula/results/statements/theorems used to solve the problems for step marking.

Q1. (a) The mean and standard deviation of a group of 100 observations are 8 and $\sqrt{10.5}$. The 50 observations selected from this group have mean 10 and standard deviation 2. Find the mean and standard deviation of the remaining 50 observations. (3 marks)

(b) Find the coefficient of variation of distribution given that mean is 120, mode is 123 and Karl Pearson's coefficient of skewness is -0.3. (1 mark)

(c) Discuss the effect of change of origin and scale on 4th moment about mean. (2 marks)

(d) Using the transformation $U = \frac{(X-32)}{2}$, for the variable X, compute first four moments about mean and hence compute the coefficients of skewness (β_1) and kurtosis (β_2) for the following frequency distribution. Also Comments on the nature of skewness and kurtosis. (6 marks)

X	22	24	26	28	30	32	34	36	38	40
f	17	26	37	45	56	74	31	47	38	29

Q2. (a) It is estimated that 10% of the population has a certain disease. A diagnostic test is available but is not perfect. There are two possible misdiagnoses. A healthy person may be misdiagnosed as sick with a probability of 5%. A person with the disease may be misdiagnosed as healthy with a probability of 1%. Given that a person picked at random is diagnosed with the disease, what is the probability that this person is actually sick? (2 marks)

(b) Let the random variable X has the probability density function

$$f(x) = \begin{cases} x, & 0 < x \leq 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the cumulative distribution function of X and hence find $P(0.3 \leq X \leq 1.5)$. Compute the mean and variance of X. (3 marks)

(c) Suppose that at two points in a room (or on a city street or in the ocean) one measures the intensity of sound caused by general background noise. Let X_1 and X_2 be random variables representing the intensity of sound at the two points. Suppose that the joint probability law of the sound intensities, X_1 and X_2 , is continuous, with the joint probability density function given by

$$f(x_1, x_2) = \begin{cases} x_1 x_2 e^{-\frac{1}{2}(x_1^2 + x_2^2)}, & x_1 > 0, x_2 > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Find the individual probability density functions of X_1 and X_2 . (ii) Further, find $P(X_1 \leq 1, X_2 \leq 1)$ and $P(X_1 + X_2 \leq 1)$. (Given $\int_0^1 x_1 e^{-x_1^2 + x_1 - \frac{1}{2}} dx_1 = 0.3592$.) (1+3 marks)

Q4. Toss a fair coin 400 times. Use the Central Limit Theorem to find an approximation for the probability of at most 190 heads. Given that $P(-\infty < Z \leq 1) = 0.8413$. (3 marks)

Q3. (a) State the additive property of Gamma distribution. (1 mark)

(b) Suppose that the number of telephone calls coming into a telephone exchange between 10 A.M. and 11 A.M. say, X_1 is a random variable with Poisson distribution with parameter 2. Similarly, the number of calls arriving between 11 A.M. and 12 noon say, X_2 has a Poisson distribution with parameter 6. If X_1 and X_2 are independent, what is the probability that more than 5 calls come in between 10 A.M. and 12 noon? (2 marks)

(c) Derive the moment generating function of Normal distribution. For the exponential distribution $f(x) = e^{-x}, x \geq 0$, find the c.d.f. of $X_{(1)}$ & $X_{(n)}$ (3+2 marks)

(d) The marks obtained by member of students for a certain subject are assumed to be normally distributed with mean 65 and standard deviation 5. If three students are taken at random from this set, what is the probability that exactly two of them have marks over 70, given $P(0 \leq Z \leq 1) = 0.3413$ (2 marks)

Prove that a continuous random variable X with having a pdf

$$f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2} e^{-x/2}, \quad x > 0$$

has a moment generating function $(1 - 2t)^{-n/2}$ where $|2t| < 1$. (2 marks)

Q4. (a) Variables X and Y have the joint pdf

$$f(x, y) = \frac{1}{3}(x + y), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 2.$$

Find:

(i) Two regression functions (2+2 marks)

(ii) Two lines of regression (1+1 marks)

(iii) $r(X, Y)$ (1 mark)

(b) From the data relating to the yield of dry bark (X_1), height (X_2) and girth (X_3) for 18 cinchona plants the following correlation coefficients were obtained: (1+1 marks)

$$r_{12} = 0.77, \quad r_{13} = 0.72, \quad r_{23} = 0.52$$

Find the partial correlation coefficient $r_{12.3}$ and multiple correlation coefficient $R_{1.23}$.

(c) Find the regression equation of X_1 on X_2 and X_3 given the following results: (3 marks)

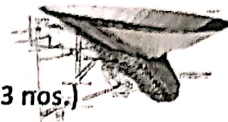
Trait	Mean	Standard Deviation	r_{12}	r_{23}	r_{31}
X_1	28.02	4.42	+0.80	-	-
X_2	4.91	1.10	-	-0.56	-
X_3	594	85	-	-	-0.40

Where X_1 = Seed per acre; X_2 = Rainfall in inches X_3 = Accumulated temperature above 42°F.

***** End *****



Indian Institute of Technology (Indian School of Mines) Dhanbad
Department of Mining Engineering
Midsem Examination Monsoon Semester 2024-2025
Course for 3rd Year (5th Semester) B.Tech. (Mining Engineering) Students (103 nos.)
Subject: Mine Planning and Economics (Code: MNC 301)



Time: 180 minutes (09:00 AM–12:00 PM) | Venue: NLHC LH 09-11 | Date: 28-11-2024 | Maximum Marks: 96

Answer all questions of the three sections and marks are assigned against each section
Focus on solving questions you know first, stay calm, manage time wisely and put forth the best effort

Section A: Numerical Answer Type Questions – 60 marks

Answer all the questions and marks are assigned against each question

1. (a) A machine costing Rs. 20,000/- was purchased and installed. Installation charges are Rs. 5,000/-. If scrap value is Rs. 5,000/- and the useful life is about 10 years, calculate the yearly depreciation by straight line method.
(b) A colliery has a total of 32 million metric tons in-situ reserves of coal. The recovery during mining is only 60%. If the production of coal from the mine is planned at 1.0 lakhs metric tons per month, estimate the life of the mine.
(c) Following sampling data from a drive is furnished:
Sample 1: 25 cm @ 3.2% Cu; **Sample 2:** 45 cm @ 2.2% Cu; **Sample 3:** 10 cm @ 3.1% Cu;
Cut-off grade is 2.0%. Minimum stoping width is 1 m. Calculate the average grade of the sample.
(d) When the nominal rate of interest is 12 % and frequency of compounding is six times a year, estimate the difference between effective rate of interest and nominal rate of interest.
(e) A mine is expected to earn Rs. 8,000/- per year for 10 years. A certain investor provided interest @ 9% per annum on the earning. What is the present value and sum of payment?
(f) Find the revenue per week that will maximize the company savings: $P = 2000 - 5Q$, where P is price per unit and Q is demand per week.
(g) Calculate the amount of stowing material/day if daily productions from a copper mine using cut-and-fill stoping is 1500 tonnes. The stowing material of 1.6 t/m³ is to be used. Ore density is 3 t/m³. Consider the value of K factor as 0.7. Use the following relation $P' = \frac{\gamma'}{\gamma} KP$; where P' = weight of stowing material to be used, P = weight of mineral (ore) to be extracted in tons. γ' = density of coal/ore in tons/m³, γ = density of stowing material in tons/m³, K = factor of stowing.
(h) Mention the relationship among, r_r = real rate of return from a mining project (%), r_i = inflation rate over the entire life of the mine (%), and r_n = nominal rate of return (%).
(i) For a shrinkage stope the following data values are given
In-situ tonnage : 9000 tonnes
In-situ grade : 5.2 g/tonne
Average grade of waste : 1.4 g/tonne
Loss of ore in the stope : 10%
Dilution : 20%
Estimate the grade at the mill-head in g/tonne?
(2 + 2 + 2 + 2 + 2 + 2 + 2 + 1 + 3 = 18 marks)
2. There is a mineral reserve of 30.8×10^6 tons containing 7.8×10^6 tons of copper ore with an average grade of 0.92%. A market survey suggested that 5000 tons of copper metal can be sold every year. Calculate daily and annual milling and mining rate, mine life, total copper recovered and overall stripping ratio. Assume mill recovery = 70%, combined smelter/refinery recovery = 96% and operating days = 300 days/year.
(4 marks)

3. Costs:

m = mining = \$1/ton
c = concentrating = \$2/ton
r = refining = \$5/lb
f = fixed cost = \$300/year

Selling Price:

s = \$25/lb

Capacities: Here tons is used for short tons (st).

M = mining = 100 tons/year
C = concentrating = 50 tons/year
R = refining = 40 lbs/year

Quantities

Q_m = amount to be mined (tons)
 Q_c = amount sent to the concentrator (tons)
 Q_r = amount of concentrator product sent for refining (lbs)

Recovery (Yield): $y = 1.0$ (100% recovery is assumed).

Table: Initial mineral inventory

Grade (lbs/ton)	Quantity (tons)
0.0-0.1	100
0.1-0.2	100
0.2-0.3	100
0.3-0.4	100
0.4-0.5	100
0.5-0.6	100
0.6-0.7	100
0.7-0.8	100
0.8-0.9	100
0.9-1.0	100
Q_m	1000

(a) Consider a combination of mining rate of 100 tons/year and concentrator cutoff grade of 0.50 lbs/ton is used and 50 tons of material having an average grade of 0.75 lbs/ton sent to the concentrator every year. The other 50 tons would be sent to the waste dump. Since the concentrator capacity C is 50 tons/year, this is an acceptable situation. The concentrator product becomes the refinery feed. In this case it would be 37.5 lbs/year (0.75 lbs/ton $\times$ 50 tons/year). Since the refinery can handle 40 lbs/year, it would be operating at below its rated capacity R. Calculate the yearly profit and NPV assuming an interest rate of 15%.

(b) Prove the following relationships to estimate the cut-off grade for maximum profit, where symbols have their usual meaning.

(i) $g_m = \frac{c}{y(s-r)}$ by constraining mining rate

(ii) $g_c = \frac{c+\frac{f}{C}}{y(s-r)}$ by constraining concentration rate

(iii) $g_r = \frac{c}{y(s-r-\frac{f}{R})}$ by constraining refining rate.

(c) After proving the above relationship for cut-off grade under different constraints, calculate the respective maximum profits and NPV assuming an interest rate of 15% using the above-mentioned details.

(d) After balancing the operations, when operations are taken in combination, the optimum/economic cutoff grades are $g_{mc} = 0.50$, $g_{cr} = 0.60$, $g_{mr} = 0.456$. Determine the overall optimum of the six cutoff grades. Perform only one iteration process to calculate the profit per year and NPV assuming an interest rate of 15% using the above-mentioned details.

(2 + 6 + 4 + 4 = 16 marks)

4.

A grade block model of molybdenum sulphide (MoS_2) has been prepared using blocks 50 ft $\times$ 50 ft $\times$ 50 ft. The tonnage factor is 12.5 ft³/st, hence, each block contains 10,000 tons. The following data were used in preparing the grade block model and in running the floating cone. The atomic weight of molybdenum (Mo) and sulfur (S) is 95.96 amu and 32.06 amu respectively.

Mining cost = \$0.74/st

Processing cost = \$1.89/st ore

General and administrative (G&A) cost = \$0.67/st ore

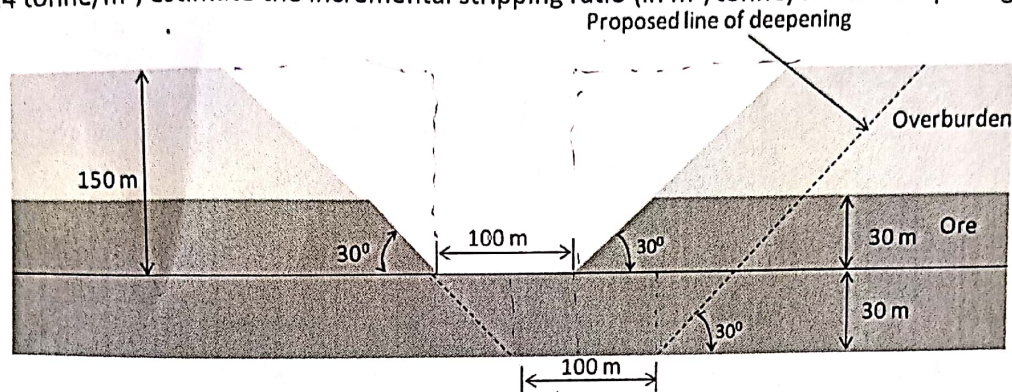
Mill recovery = 90%

Selling price = \$6/lb contained molybdenum (F.O.B. mill site)

Calculate the breakeven grade (%) for MoS_2 and also, compare the breakeven analysis for a price of \$2/lb and \$1.5/lb molybdenum.

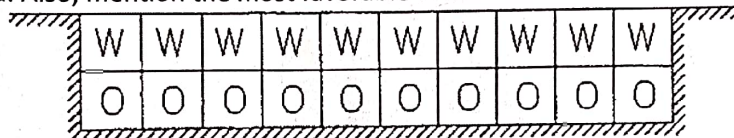
(4 marks)

5. An operating surface mine is proposed to be deepened by 30 m as shown in the figure. If the density of the ore is 2.4 tonne/m³, estimate the incremental stripping ratio (in m³/tonne) for the deepening,



(3 marks)

6. Consider the property shown in the figure below which has 10 ore blocks overlain by 10 waste blocks. A production rate of 4 blocks per year (irrespective of whether the blocks are ore or waste) will be assumed. The net value for an ore block is Rs. 2 and the cost of removing the waste is Rs. 1/block. The total cost involved in waste removal would be Rs. 10 and the ore value is Rs. 20. If both ore and waste could be mined instantaneously, the net present value would be Rs. 10. However, it is not possible to extract all of these 20 blocks at once due to practical constraints. Discuss any four combinations of scheduling scenarios which may be considered. Also, mention the most favorable combination out of them based on NPV (@10%).



(4 marks)

7. An economic block model is shown below.

(a) Show the final pit layout using Floating Cone Technique.

(b) Determine the sequence in which the blocks should be mined as per the algorithm developed by Roman (1974).

①	-1	-1	-1	-1	-1	2	1	1	1	-1	-1	-1	-5	+5	⑥
②	-1	-1	-2	-2	-2	1	2	3	2	1	-2	-1	-4	+9	⑤
③	-2	-1	-1	-1	2	4	5	3	2	1	-2	-2	16	⑩	31
④	-4	-3	-2	-2	-2	1	3	4	2	-1	-1	-2	8	④	2
⑤	-6	-6	-5	-5	-4	-5	2	3	-2	-2	-3	-4	-5	②	
⑥	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6		

(2 + 2 = 4 marks)

8. Using the table shown below, determine

(a) the accounting rate of return for the initial capital invested and life of the project is 4 years.

(b) dynamic payback period using linear interpolation for discount rate of 8% per annum.

Time	0	1	2	3	4
Net Cash Flows	-100,000	28,000	30,000	35,000	32,000

(2 + 2 = 4 marks)

9. A mill concentrates, having 25% copper, is proposed to be sold at Rs. 1,25,000 per tonne. The grade of the deposit is 0.8% Cu and the overall cost of mining and milling is Rs. 2,520 per tonne of ore. At a recovery of 75%, estimate profit (in Rs./tonne) of concentrate?

(3 marks)

Section B: Descriptive Answer Type Question (Answer in 2-3 lines/pointwise) – 15 marks

Answer all the questions and marks are assigned against each question

10. Name the stoping method/methods with following features:
a. Costliest b. Cheapest c. Highest dilution d. Least recovery e. Highest productivity f. Lowest productivity
g. Early production but highest development h. Subsidence unavoidable i. Ore tie up during stoping
j. Least exposure to unsafe conditions. (5 marks)
11. Differentiate
(a) Salting and Sampling (b) Overhand and underhand stoping (c) Elastic demand and inelastic demand
(d) Escrow amount of coal/lignite and mineral (e) Major minerals and minor minerals
(f) Project year, Production year and Calendar year (6 marks)
12. Present a layout of cut and fill stope. Write down suitable conditions for its application and variants. What are the advantages and limitations. Mention the range of dimension used to space the levels. (2 marks)
13. Present a layout of sublevel stoping. Write down suitable conditions for its application and variants. What are the advantages and limitations. Mention the range of dimension used to space the levels. (2 marks)

Section C: Descriptive Answer Type Question (Answer in 2-3 lines/pointwise) – 21 marks

Answer all the questions and each question carries 1 mark only

14. Mention the basic line items in a pre-production cash flow table.
15. Mention the basic line items in a production period cash flow table.
16. Taylor has provided some useful insights into deciding the appropriate mine life. Summarize the factors which should be considered. Develop the reasoning as to why one might expect the life of a mine to be related to the cubic root of the ore tonnage.
17. Discuss UNFC reserve classification system.
18. Discuss the objectives and key provisions of National Mineral Policy 2019.
19. Mention the purpose of mine valuation.
20. Discuss the methods of investment appraisal.
21. Mention different kinds of mineral price quotation and factors affecting mineral price.
22. Discuss the steps taken by Indian government to ease the process of licensing, clearances and approvals required for operating a mine.
23. Discuss methods of mineral conservation for sustainable development.
24. Discuss coal mine closure plan.
25. Briefly describe the various stope selection techniques and their applicability.
26. What are the assumptions that need to be made in the mine life-plant size determination?
27. Group (a) orebodies, country rocks based on their stability and (b) orebodies based on their thickness and dip
28. List factors governing choice of a mining method.
29. List the desirable features a stoping method should be able to render, if selected.
30. Classify underground stoping methods.
31. Discuss model and design parameters considered while designing stope.
32. Discuss the different pillar types during different stoping methods.
33. Discuss ground control issues associated with different stoping methods.
34. Discuss liquidation and their techniques.

Best Wishes!

Question Paper Ends Here